

A concrete example of a metabolomics study in health

Development, validation, transfer to clinic of a blood metabolic biomarker for early diagnosis of Parkinson's disease

Grenoble Institute of Neurosciences (GIN)

Sabrina Boulet, Florence Fauvelle



A concrete example of a metabolomics study in health

Plan

- **Parkinson's disease and the current challenges**
- **Our translationnal study for biomaker development**
- **Toward clinical application of our biomarker**
- **Lesson learned, perspectives**

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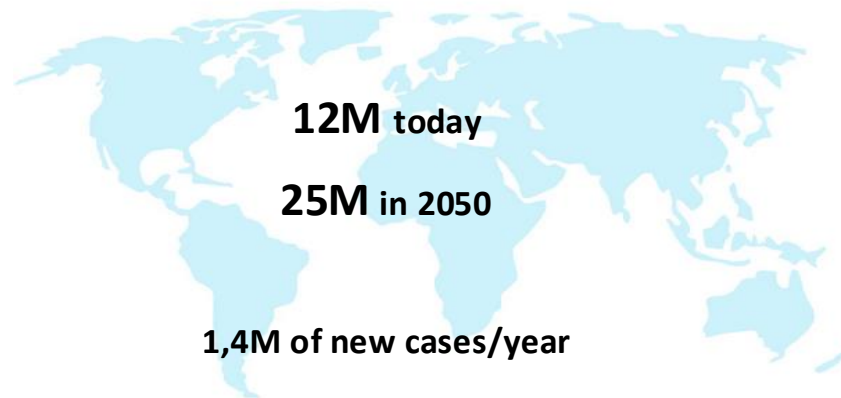
Parkinson's Disease (PD)

- An incurable neurodegenerative disease in constant progression in the world independently of general population aging
- A major cause of handicap

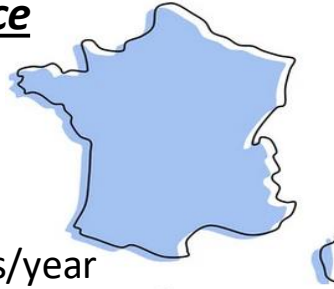
Parkinson's Disease Symptoms



In the world



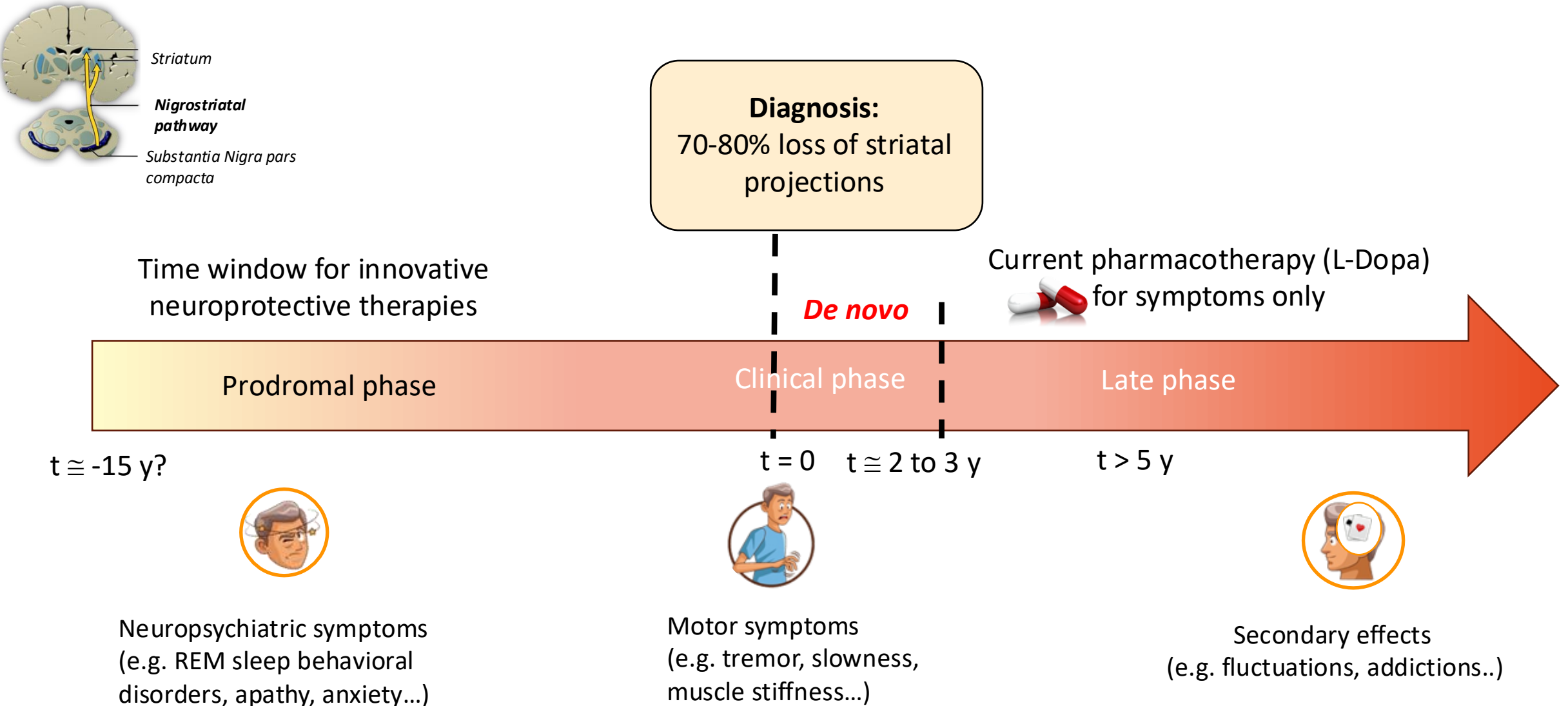
In France

- 
- 270 000 cases
 - 27000 new cases/year
 - 1 person on 50

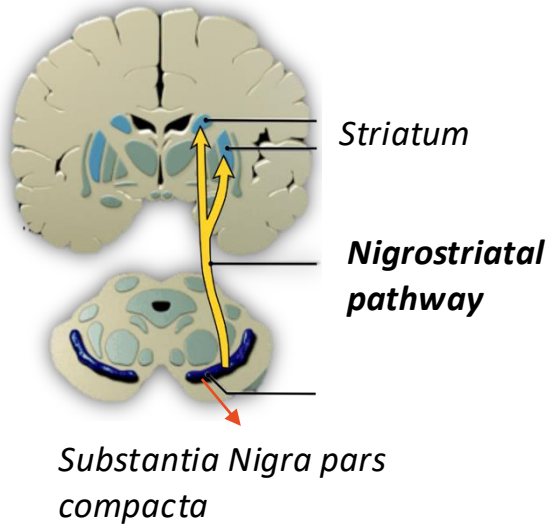


Case number x 2 by 2050

Parkinson's Disease (PD) phases



Parkinson's Disease (PD) and current challenges



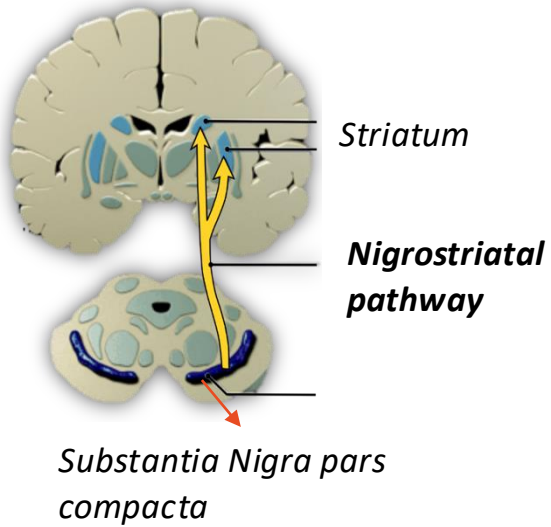
CHALLENGES

Better **understand** the underlying **mechanisms** of the disease

Improve the performance and reliability of **current diagnostics**

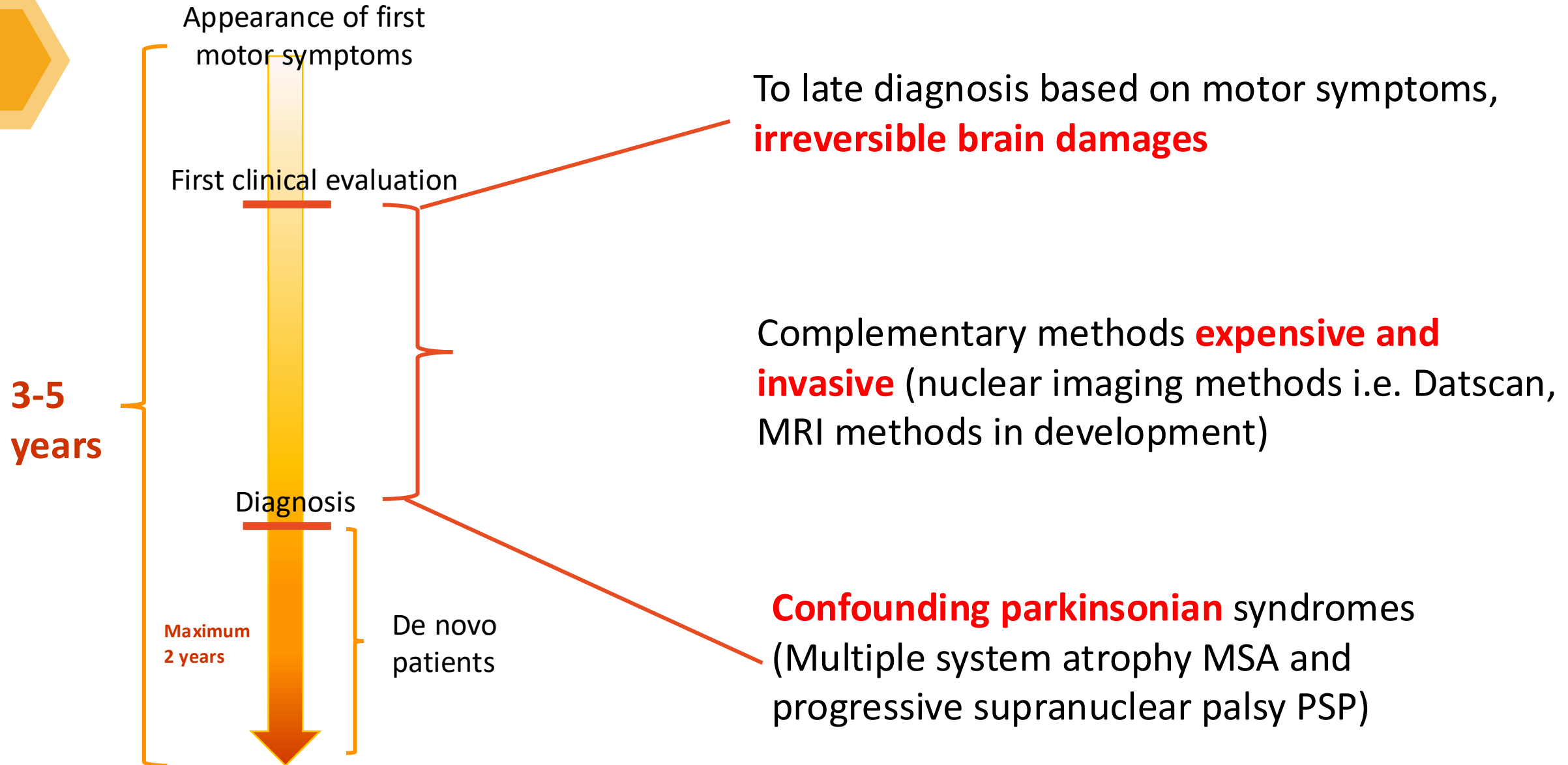
Develop **early diagnosis** methods

Parkinson's Disease (PD) and current challenges

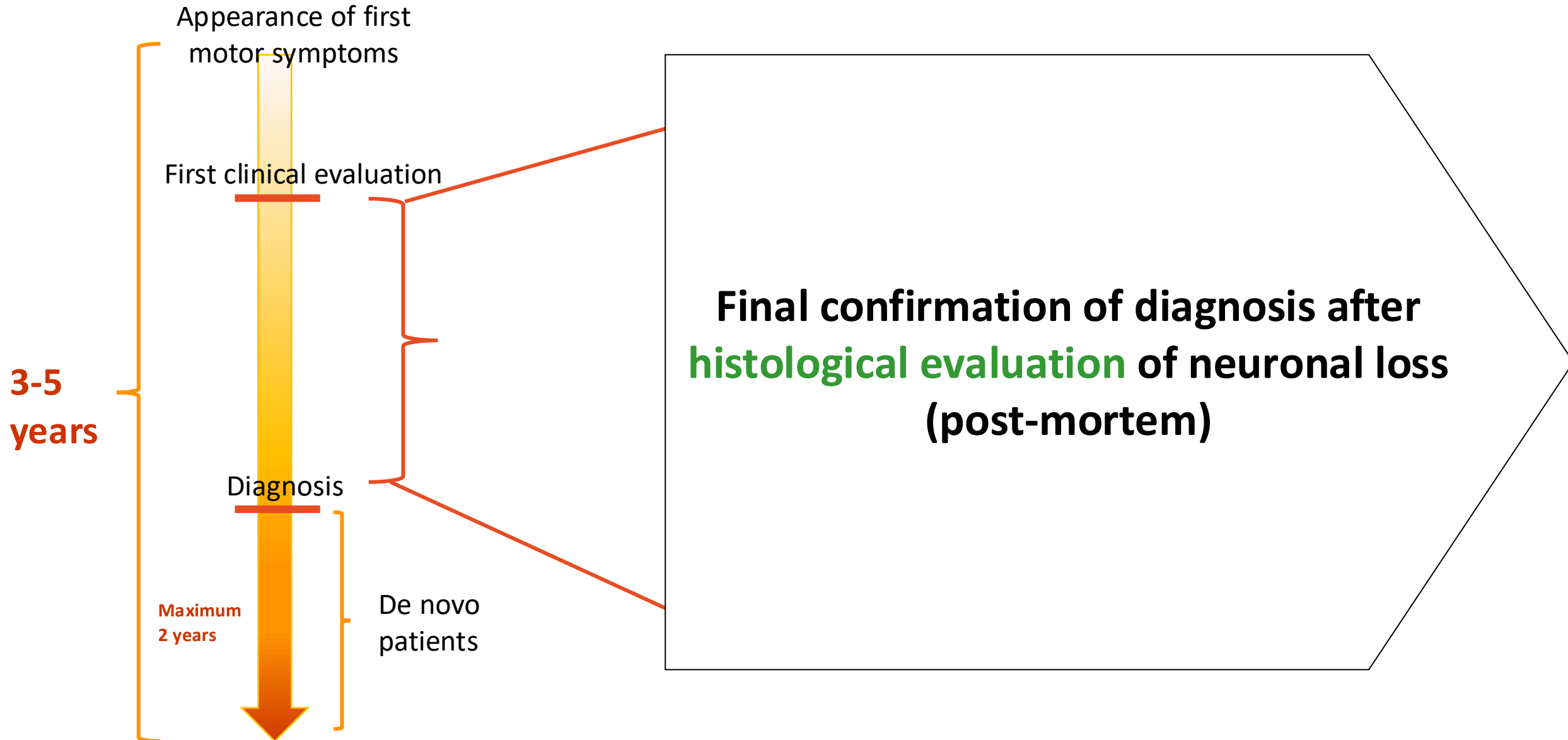


- Enable the development/testing
- of new therapies before neuronal damage is too far advanced
- Allow building of reliable and homogenous biobanks of prodromal patients

Improve the performances of current diagnosis



Improve the performances of current diagnosis



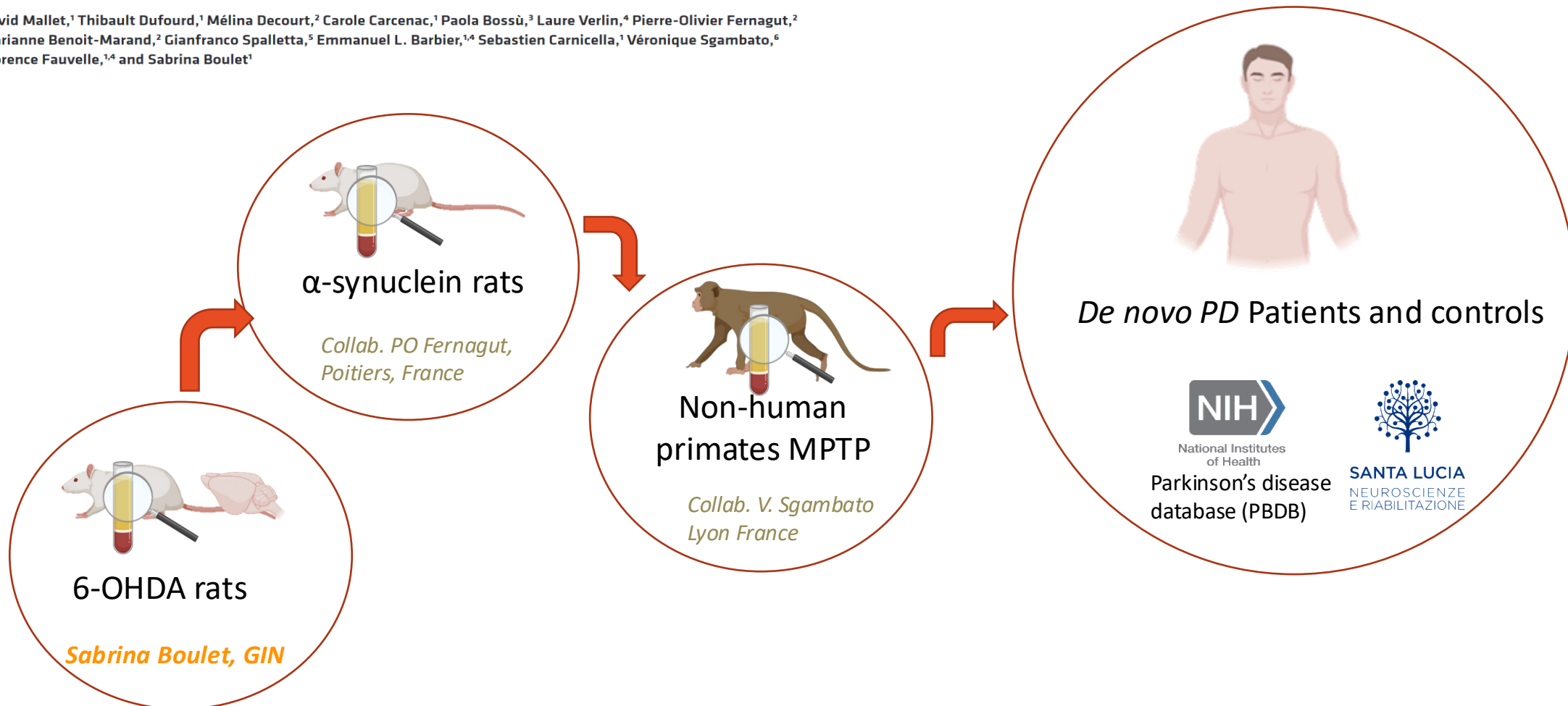
Translational study

The Journal of Clinical Investigation

CLINICAL MEDICINE

A metabolic biomarker predicts Parkinson's disease at the early stages in patients and animal models

David Mallet,¹ Thibault Dufourd,¹ Mélina Decourt,² Carole Carcenac,¹ Paola Bossù,³ Laure Verlin,⁴ Pierre-Olivier Fernagut,² Marianne Benoit-Marand,² Gianfranco Spalletta,⁵ Emmanuel L. Barbier,^{1,4} Sebastien Carnicella,¹ Véronique Sgambato,⁶ Florence Fauvelle,^{1,4} and Sabrina Boulet¹

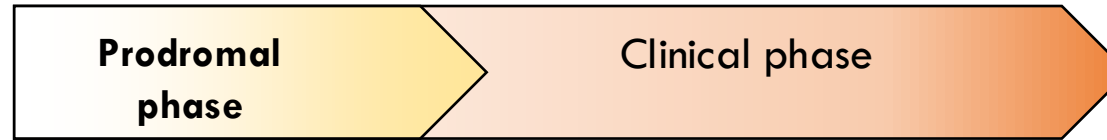


Disease progression in 6-OHDA rat model



6-OHDA* rats

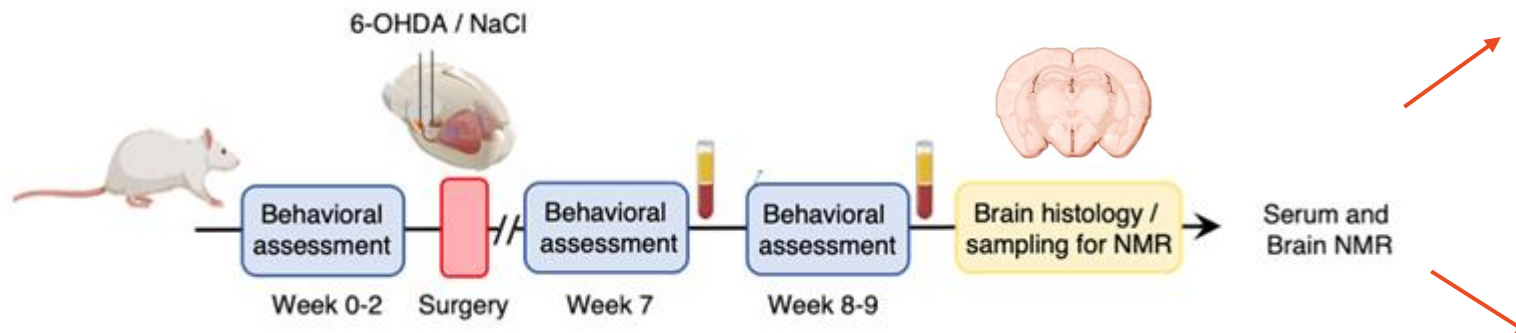
Non progressive model



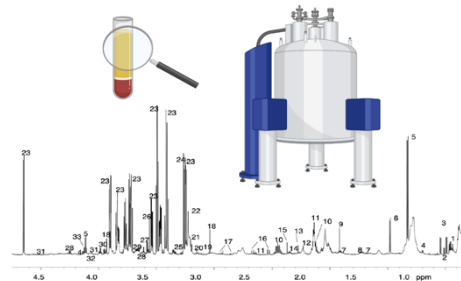
Loss of motivation

Motor symptoms

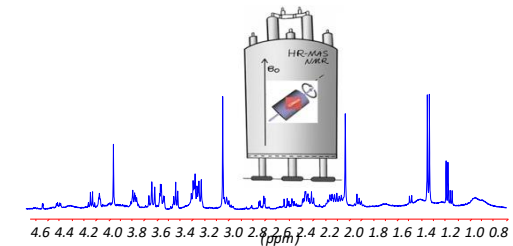
Behavioral disturbances



Immunohistology (Tyrosine-hydroxylase)



^1H -NMR, 950 MHz (Infranalytics, IBS-Grenoble),
raw serum diluted with PBS/D₂O

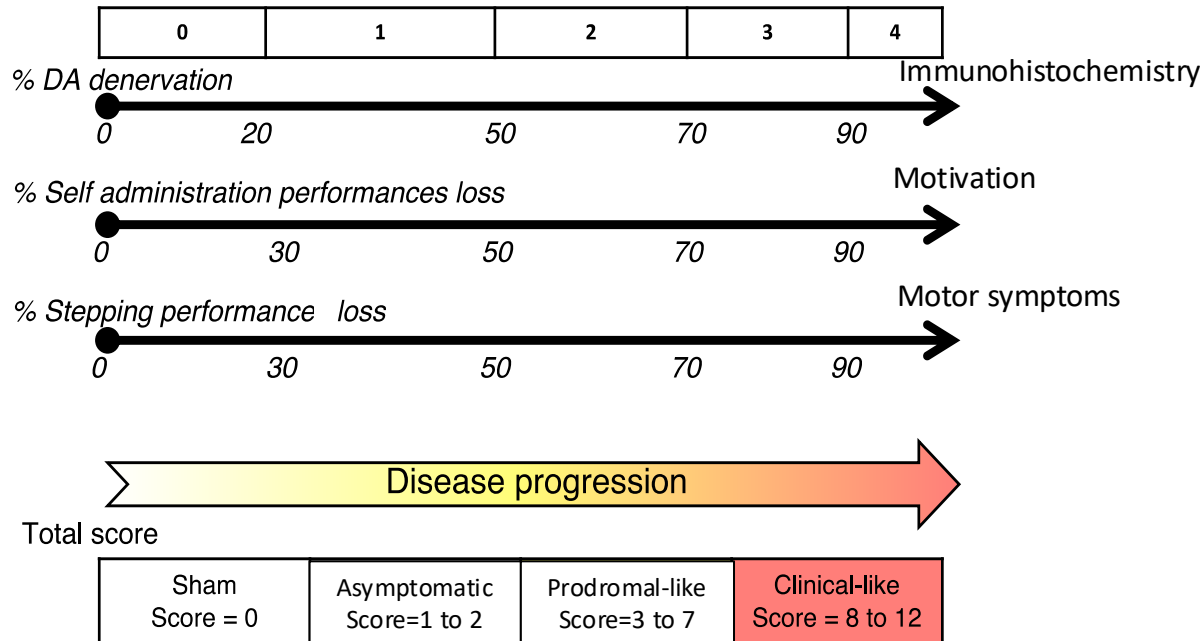


^1H -HRMAS NMR, 500 MHz, for
intact brain tissue (CEA-Grenoble)

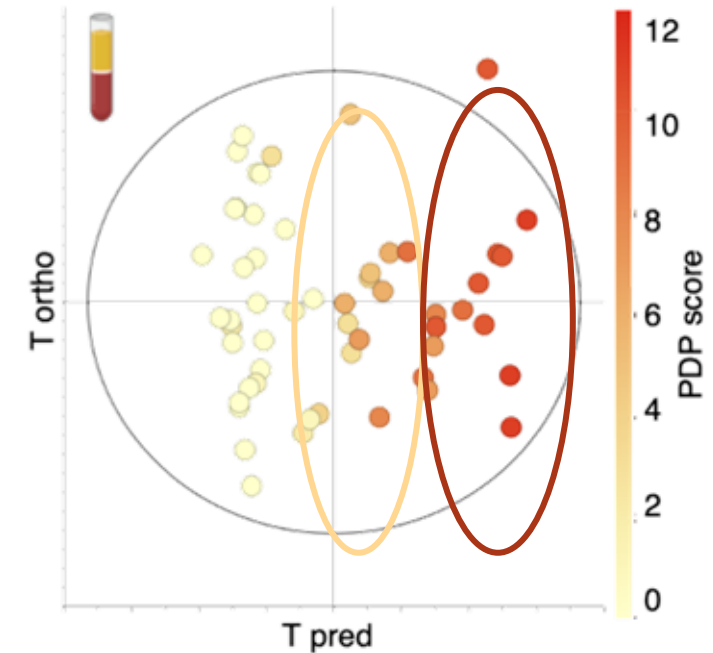
* 6-OHDA: 6-hydroxydopamine

Disease progression in 6-OHDA rat model

Score of PD progression (PDP score)

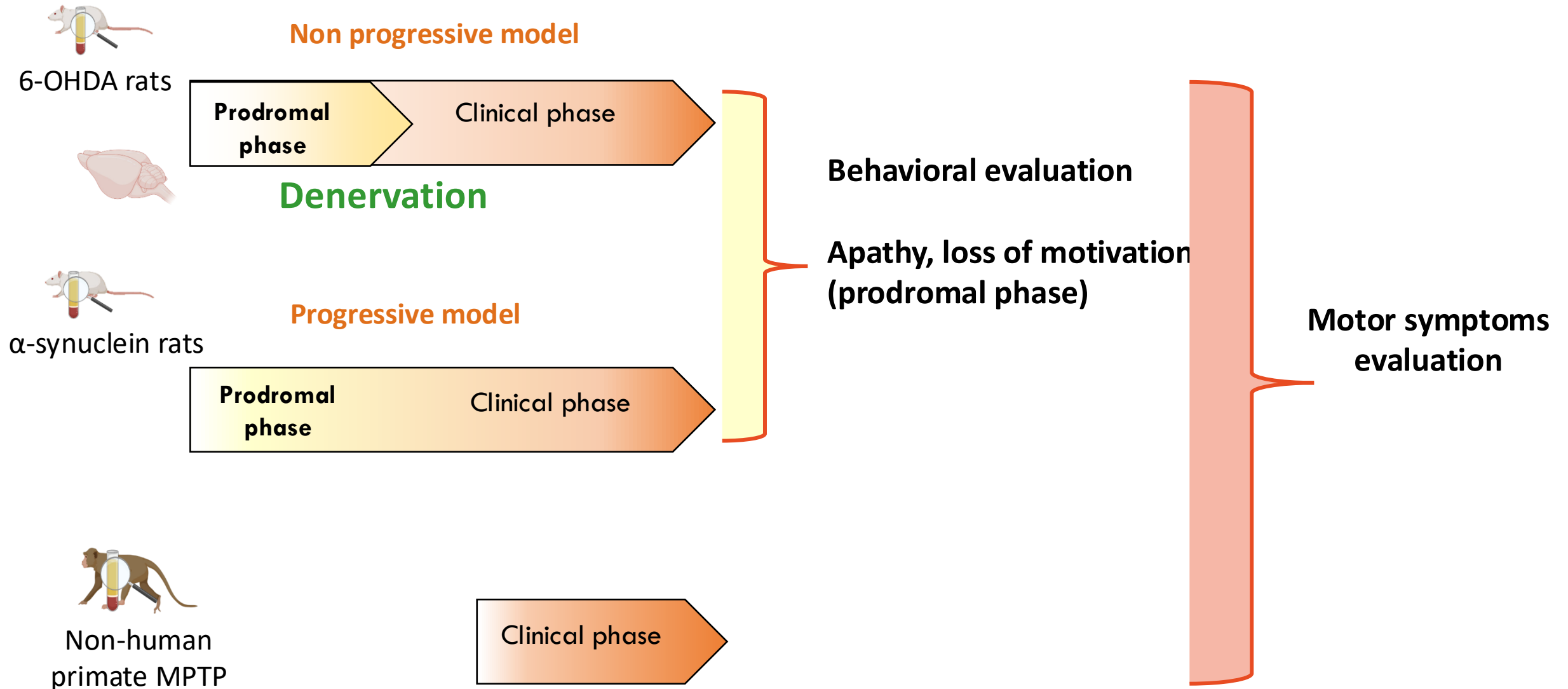


6-OHDA serum OPLS model



- Variation in metabolic profile correlated with disease progression
- Metabolic dysregulations observed as soon as the prodromal phase

Translational study: animal models



Blood metabolic biomarker development

NMR analysis of Serum samples

Metabolic profile in each cohort



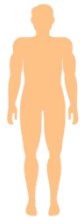
*Serum in fasting animals
Applying to biobanks*



*Multivariate statistics:
OPLS, OPLS-DA*



Non-human primates MPTP



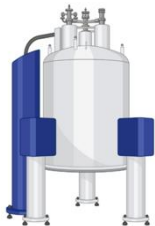
De novo PD patients
(USA, Italy)



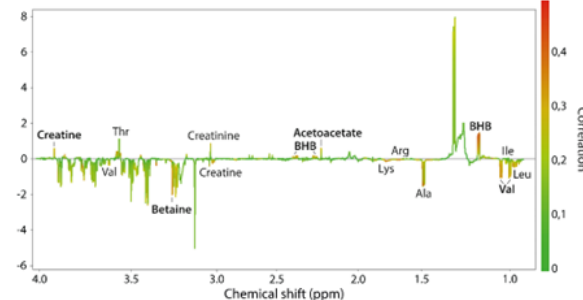
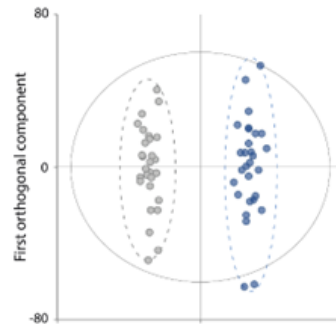
α -synuclein rats



6-OHDA rats



^1H -NMR, 950 MHz



Blood metabolic biomarker development

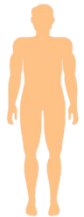
NMR analysis of Serum samples



*Serum in fasting animals
Applying to biobanks*



Non-human
primates MPTP



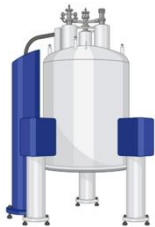
De novo PD patients
(USA, Italy)



α -synuclein rats



6-OHDA rats

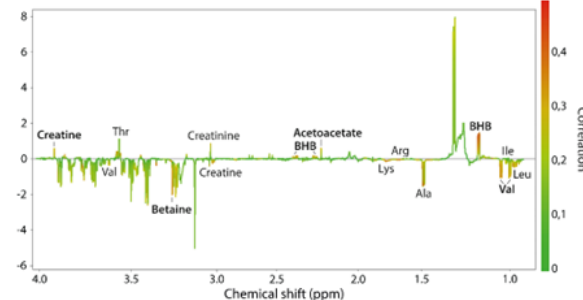
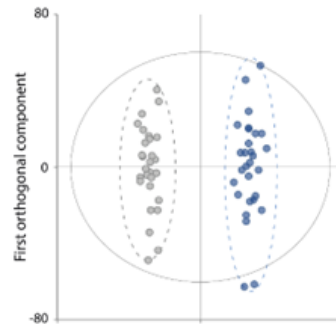


^1H -NMR, 950 MHz

Metabolic profile
in each cohort



*Multivariate statistics:
OPLS, OPLS-DA*



Selection of most discriminant
metabolites + Metabolites
dysregulated in 3 of 4 species



*Multivariate and
univariate statistics*



**Acetoacetate
betaine
Beta-hydroxybutyrate
creatine
pyruvate
valine**

Blood metabolic biomarker development

NMR analysis of Serum samples

Metabolic profile in each cohort

Selection of most discriminant metabolites + Metabolites dysregulated in 3 of 4 species

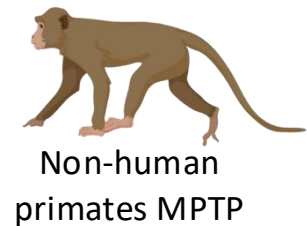
Biomarker building and characterization

*Serum in fasting animals
Applying to biobanks*

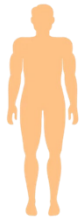
*Multivariate statistics:
OPLS, OPLS-DA*

*Multivariate and
univariate statistics*

*Logistic regression, ROC
curve, AUC, accuracy,
external validation*



Non-human primates MPTP



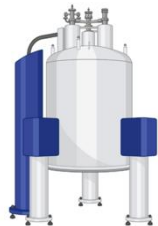
De novo PD patients
(USA, Italy)



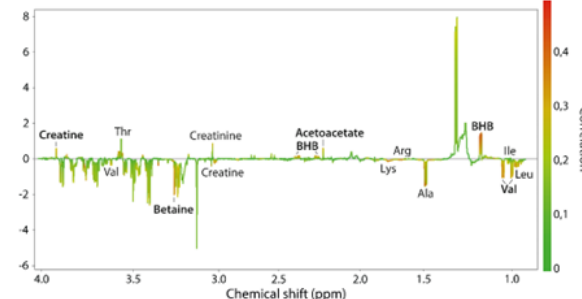
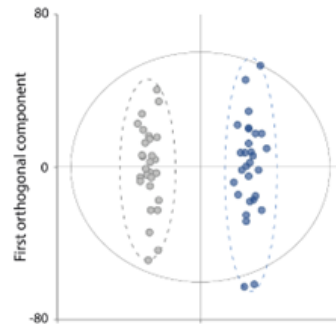
α -synuclein rats



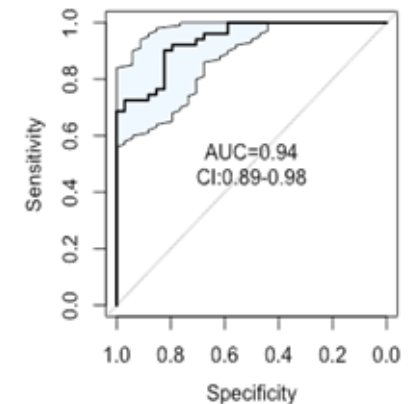
6-OHDA rats



^1H -NMR, 950 MHz

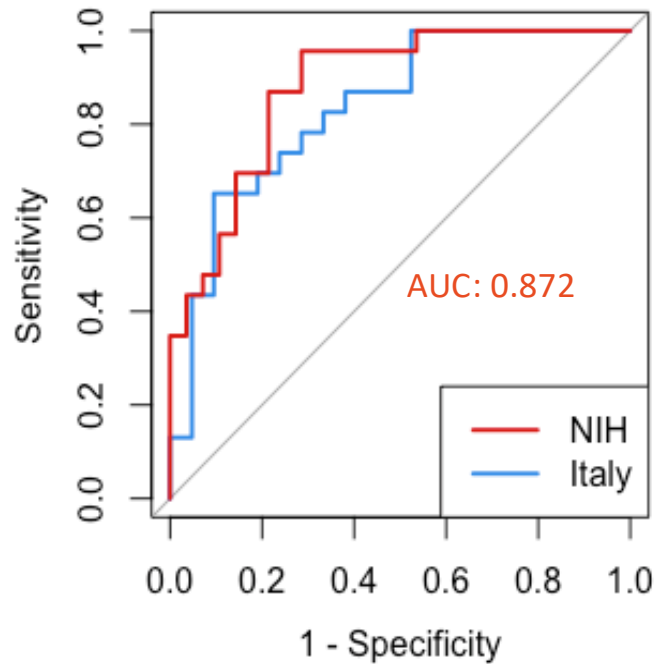


**Acetoacetate
betaine
Beta-hydroxybutyrate
creatine
pyruvate
valine**



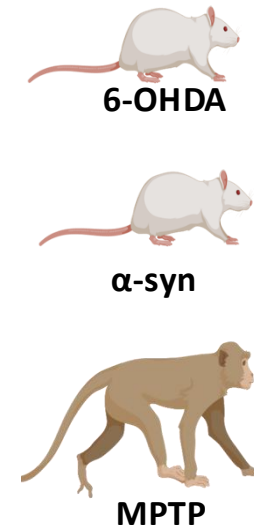
Accuracy 85 %

A new blood metabolic biomarker for PD

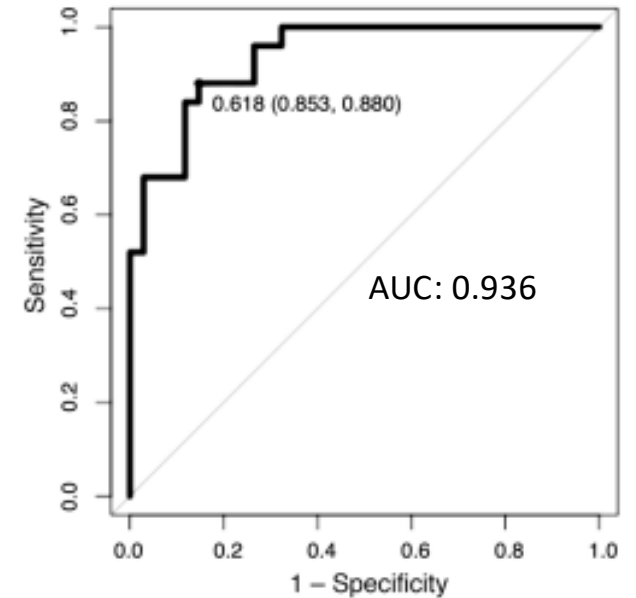


Accuracy 85 %

n=51; ROC curve
n=36; external validation



n=61; ROC curve
n=40; external validation



Accuracy 92,5%

Model built with « de novo-like » PD animals

Prodromal-like ones are predicted with 92,5 accuracy

Towards clinical application of the biomarker

Evaluation of the biomarker performances

At de novo
stage

Evaluate performances for **differential diagnosis** against confounding parkinsonian syndromes

Evolution of
pathology

Correlation with clinical scores and **disease evolution**, prediction of worsening, fluctuation ...

Predict PD
earlier

Prediction of **PD phenoconversion** in prodromal individuals

Transfert to high throughput metabolomics

Clinical
application

Bruker IVDr 600 MHz and B.I.QUANT PS for small molecules/B.I.LISA for lipoproteins

Towards clinical application of the biomarker

Evaluation of the biomarker performances

At de novo
stage

Evaluate performances for **differential diagnosis** against confounding parkinsonian syndromes

Transfert to high throughput metabolomics

Clinical
application

Bruker IVDr 600 MHz and B.I.QUANT PS for small molecules/B.I.LISA for lipoproteins

Differential diagnosis for PD



Parkinson's disease
database (PDBD),
new samples

Neurodegenerative
diseases

Parkinsonian
syndrome

Alzheimer disease

n=33



S-Park project

Parkinson's disease
(PD)

de novo PD patients (n=30)
Homologous controls (n=30)

Atypical parkinsonian
syndromes

Multiple System Atrophy
(MSA)

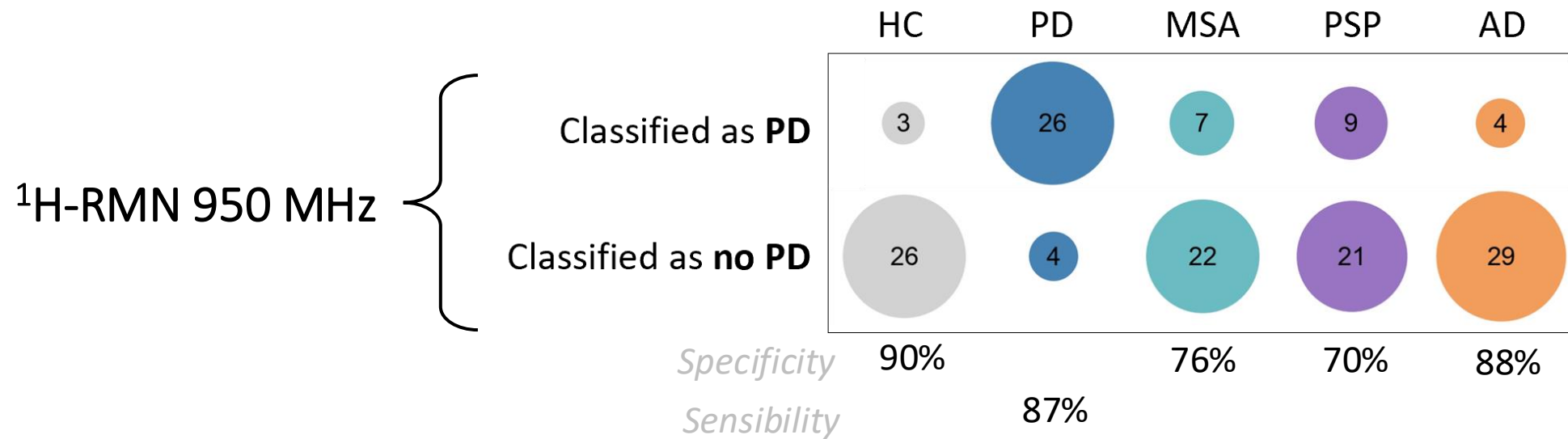
n=30

Progressive Supranuclear
Palsy (PSP)

n=30

Others (DCL, DCB)

Differential diagnosis of PD



HC homologous controls

Biomarker performance at with the IVDr 600 MHz

IVDr specifications

5 spectra acquired for each sample

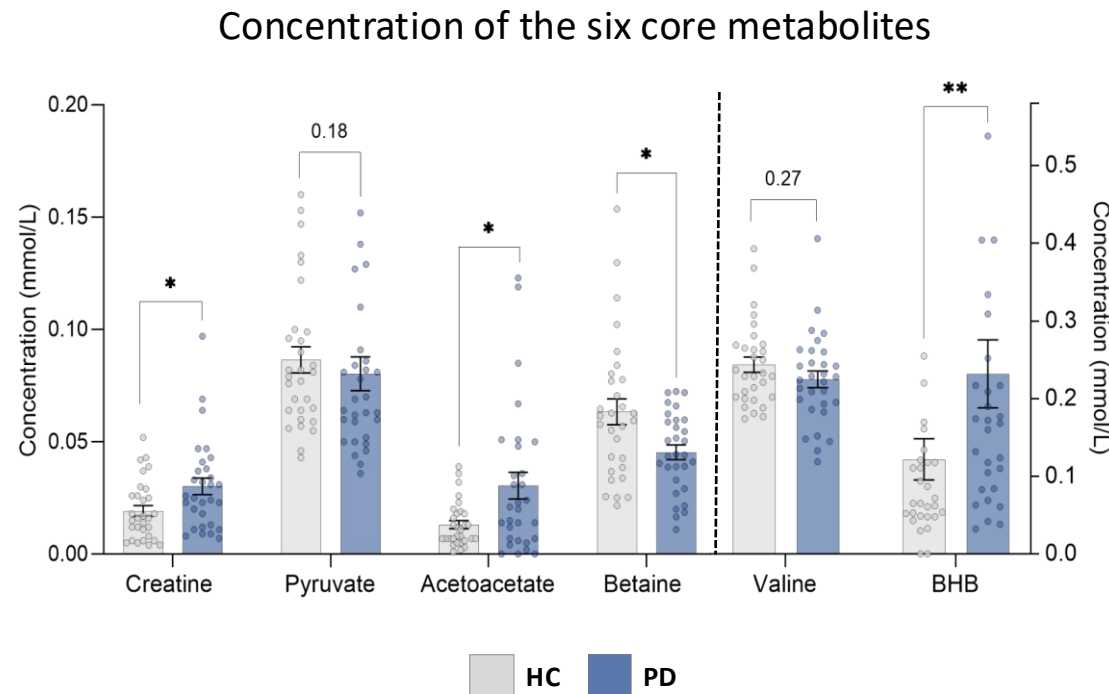
5 of the 6 metabolites quantified by Bruker in the NOESY1D (?) using B.I.QUANT-PS2.0

Betaine was not quantified as it is not detected in 1D spectra at 600 Mhz (overlap with TMAO and with glucose)

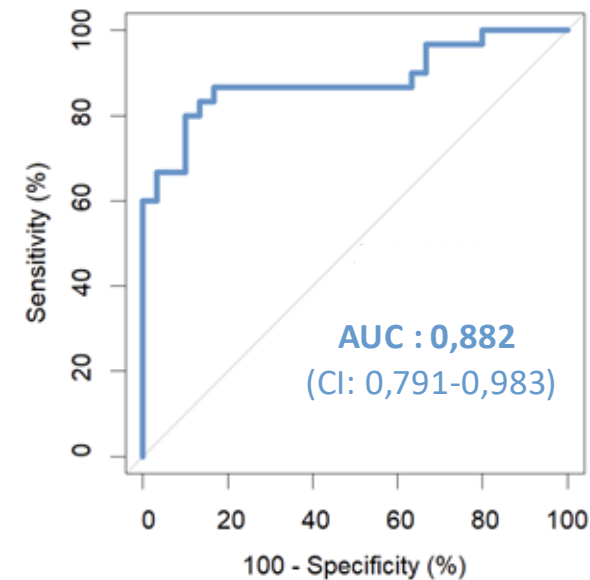
Bruker* developed the method, from spiking experiments (CH3) and from 2D-Jres spectra (CH2) (not yet included in the B.I.QUANT-PS2.0 method)

*Nils Pompe and Claire Cannet

Biomarker performance at 600 MHz



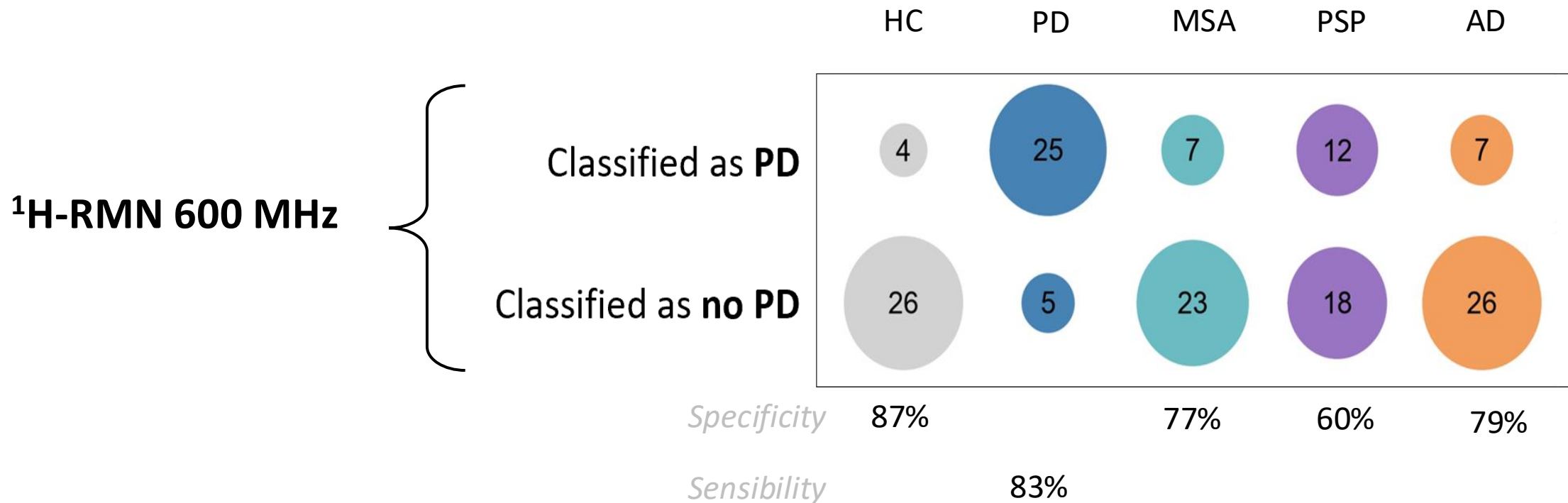
ROC curve, PD/HC cases



DeLong's test for two ROC curves data:
ROC 950MHz (AUC: 90.34)/
ROC 600MHz (AUC: 88.22)
p-value = 0.7399

Slight but *non significant* loss of performance at 600 MHz

Differential diagnosis of PD at 600 MHz



Differential diagnosis of PD at 600 MHz

- **Improve performances at 600 MHz**

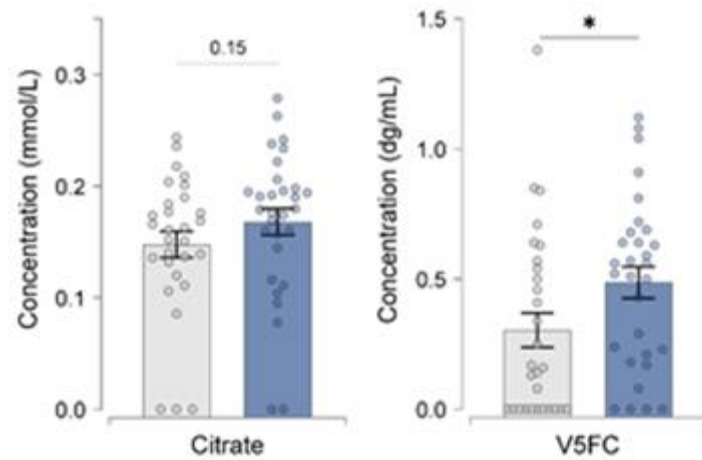
41 metabolites quantified using B.I.QUANT-PS2.0 and 112 lipoproteins were quantified using B.I.LISA packages

- Selection of 7 additional variables using LASSO (while excluding the 6 core metabolites)
- 2 then 3 out of 7 retained based on:
 - One by one addition of variable to the 6 core metabolites model
 - Biological relevance
 - PD/HD mean comparison and univariate statistics
 - OPLS-DA and VIP

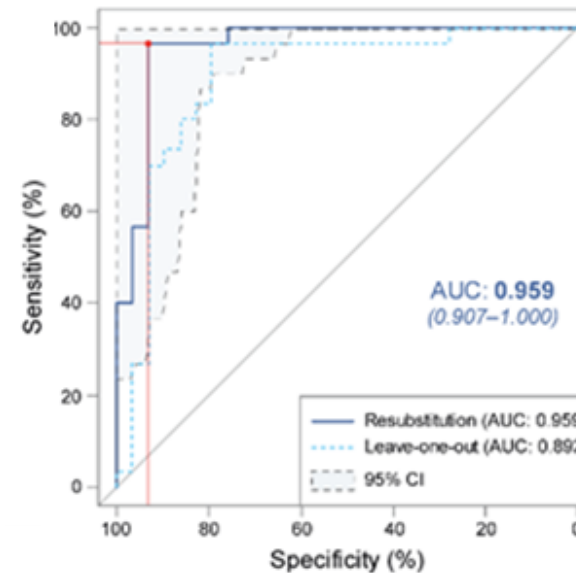
Model performances assessed through ROC curves (AUC, sensitivity, specificity, PPV, NPV, accuracy,) (Leave-One-Out cross validation for AUC)

Differential diagnosis of PD at 600 MHz

- Improved performances at 600 MHz



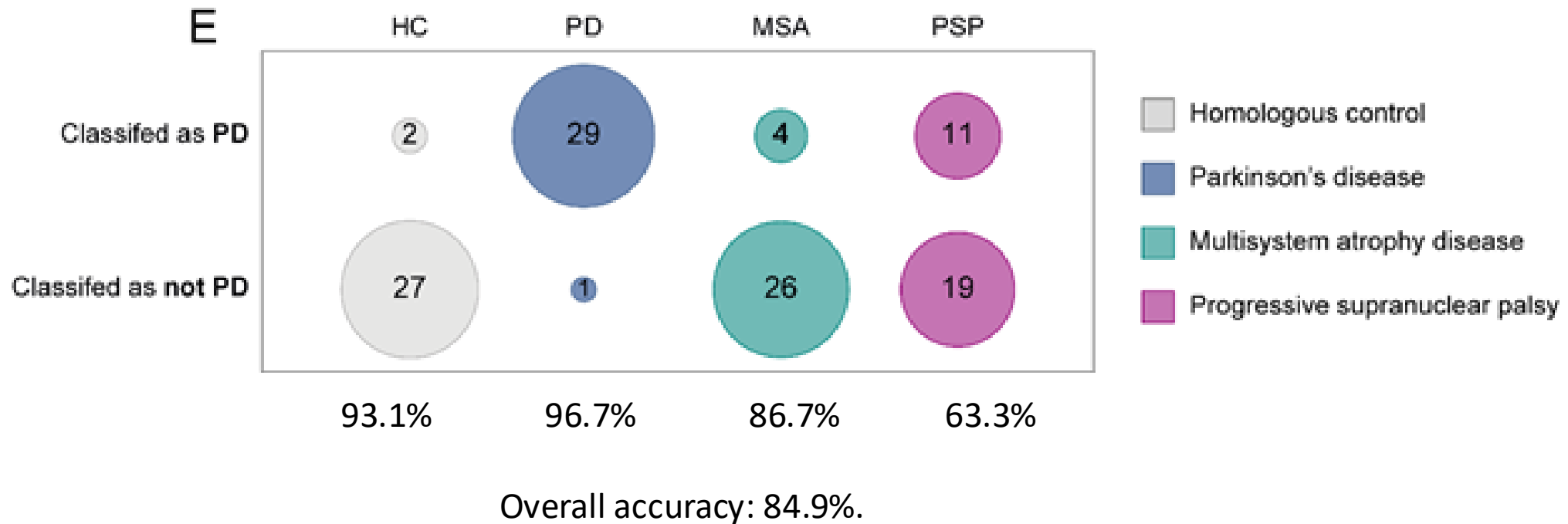
V5FC: VLDL-5 free cholesterol



At the threshold
96.7% sensitivity
95% specificity

Differential diagnosis of PD at 600 MHz

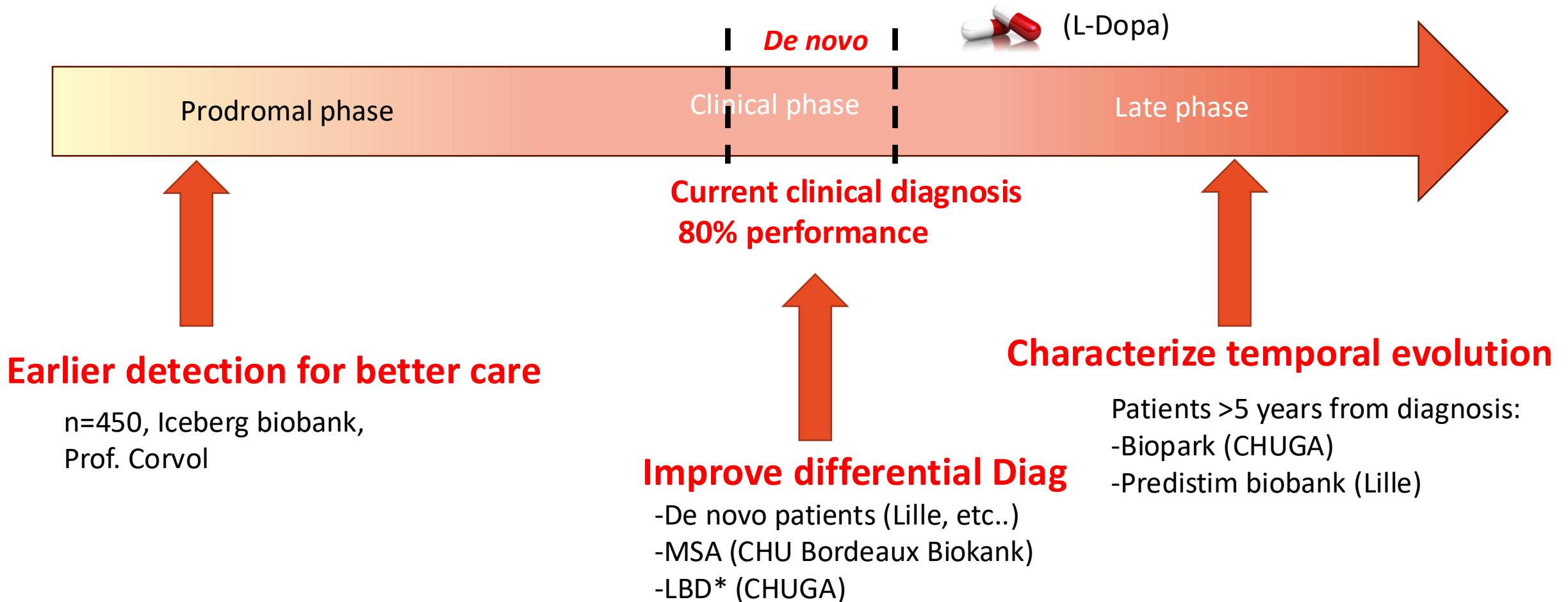
- Improved performances at 600 MHz



The Park-in-sang maturation project

A project granted by the SATT Linksium (Société d'accélération de transfert de technologie)

What clinicians/private neurologists/physicians want:

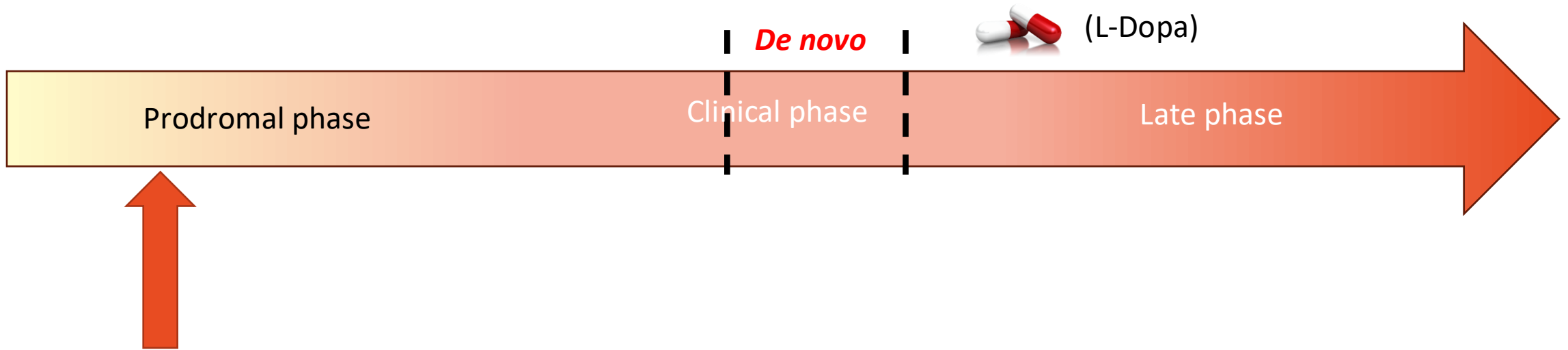


*Lewy Body Dementia

The Park-in-sang maturation project

A project granted by the SATT Linksium (Société d'accélération de transfert de technologie)

What pharmaceutical industry wants (and clinicians):



Earlier detection

A compagnon test for drug development:

- To better stratify patients, evaluate new treatments and their efficiency in targeted population
- Develop targeted therapies

Summary

Our experimental strategy was very powerful:

- **High degree of diversity** in terms of:
 - species studied (rats, monkey, humans),
 - origin of human samples (USA and Italy),
 - PD induction: 2 different toxins, alpha-synuclein-AAF models
- But: **high degree of homogeneity** of cohorts (« **true** » **de novo** patients, rare in the literature (<2 years, untreated)
- Biomarker **only based on diagnosis**: PD, not PD, excluding clinical and demographic parameters (sex, age, geographical origin) etc..
- External validation of the biomarker

Practically

- High magnetic field allowed to identify metabolite (eg betaine) not detected at lower field
- Relative quantification and absolute one lead to the same results –except for valine
- 2 magnetic fields for at least 500 human samples is very rich

AT GIN: there is a **close proximity between researchers, clinicians** (who are part of the teams), **different animal models** are developed and are available

Toward clinical research: a different job

- pb of clinician availability: cohort size evaluation can be touchy..
- RIPH Clinical trials protocols: around 50 pages of specific language and administrative parts*
- Researchers are not PI of the project, even if they initiated it
- A clinical trial is expensive (mainly when recruiting patients: insurance, monitoring, sample collection etc..) (we paid around 50KE to recruit 70 PD patients during their routine check-up visit, with only a blood sampling)
- Even when applying to biobanks of collaborators: long process, sometimes free, sometimes not

Clinical research is very time consuming, requires a lot of determination and perseverance

Diagnosis: a very challenging domain

- Metabolomics is a new method not completely « established » « standardized » « known » by clinicians
- NMR is not yet a method used in clinic for biological analyses at hospital (unless mass spectrometry), but there are few companies in the world developing NMR-based targeted methods
- Big corporations in genetic and proteins analysis (including pathological proteins like alpha-syn in Parkinson) are capturing the market

More than improving diagnosis, there is a need of companion test for new neuroprotective therapies developed by big pharma

Team Physiopathology of motivation

Sébastien Carnicella
Sabrina Boulet

Team Functional neuroimaging and brain perfusion

Emmanuel Barbier
Florence Fauvelle

David Mallet (past PhD student)
Vanille Millasseau (PhD student)
Eda Nur Kavlak (Park-in-sang project)

CEA

Pierre Alain Bayle
Michel Bardet

CHUGA

Audrey Le Gouellec
Andrea Kistner
Elena Moro
Mathilde Sauvee

IBS

Adrien Favier
Alicia Vallet

Gemeli

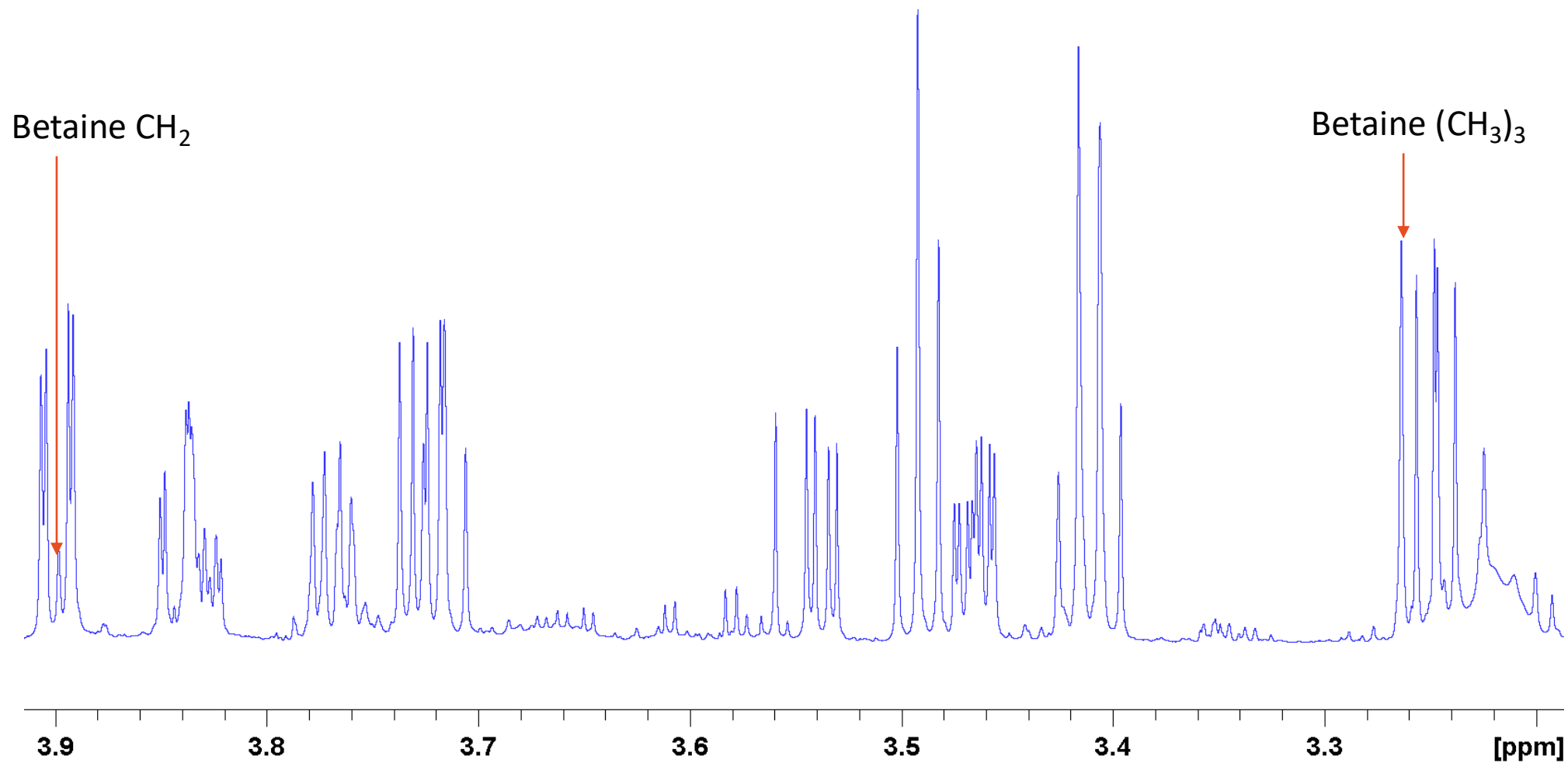
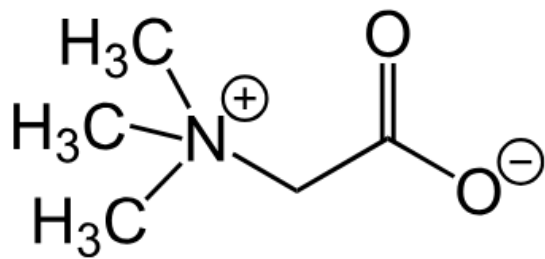
Bénédicte Elena-Herrmann
Joran Villaret

Bruker Biospin

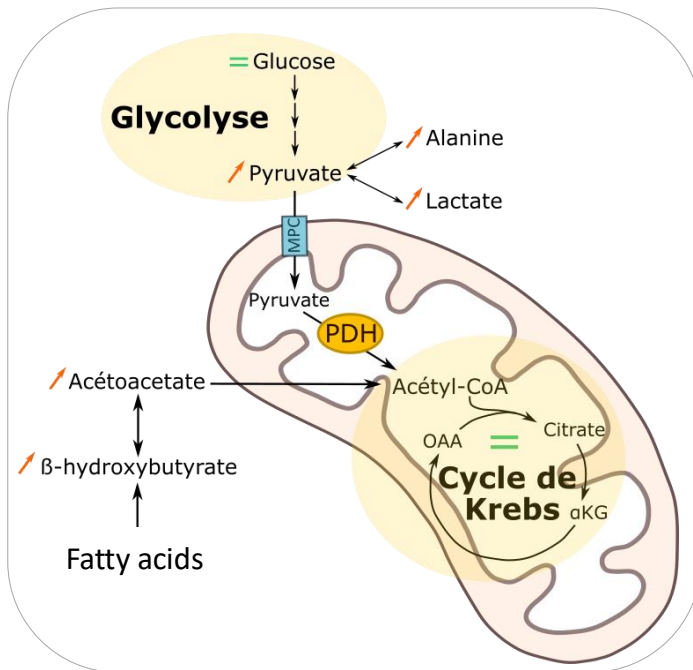
Nils Poms
Claire Cannet



Merci de votre attention



Parkinson's Disease (PD) and pyruvate metabolism



Decoupling between glycolysis and Krebs cycle



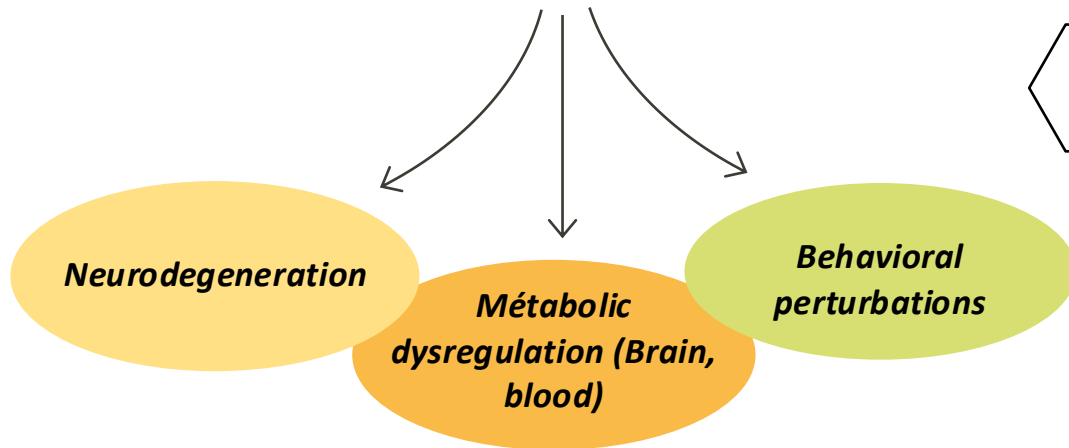
Role of Pyruvate Deshydrogenase (PDH) ?

Preclinical projet:

Understand the role of PDH in PD physiopathology

Parkinson's Disease (PD) and pyruvate metabolism

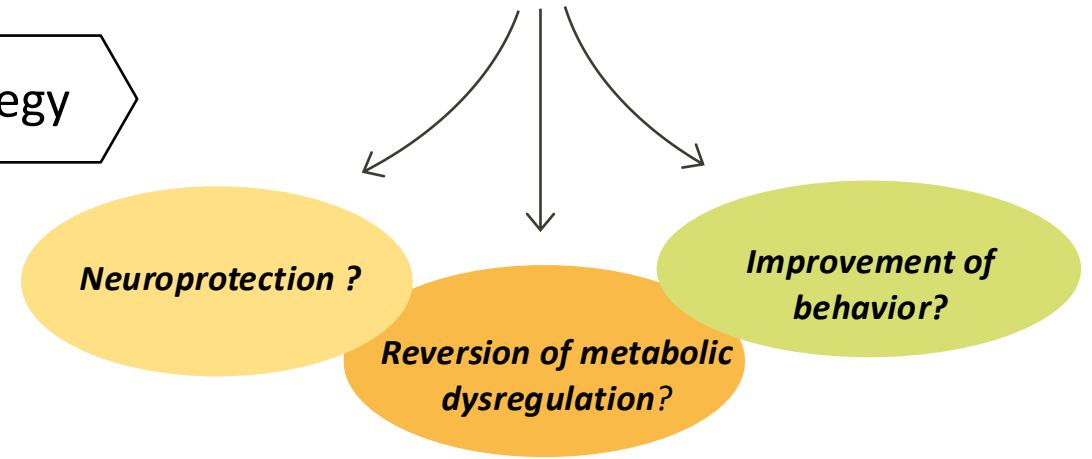
PDH down-regulation in healthy rats



Parkinsonian phenotype at the prodromal stage

PDH up-regulation in PD rat models

AAV strategy



Protection against PD ?

Work in progress